



# basic education

Department:  
Basic Education  
**REPUBLIC OF SOUTH AFRICA**

## SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

### ENGINEERING GRAPHICS AND DESIGN P2

MAY/JUNE 2025

MARKS: 100

TIME: 3 hours

This question paper consists of 6 pages.

Barcode label

DO NOT FOLD THE QUESTION PAPER IN HALF.

## INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions.
2. Answer ALL the questions.
3. ALL drawings are in third-angle orthographic projection, unless otherwise stated.
4. ALL drawings must be prepared using pencil and instruments, unless otherwise stated.
5. ALL answers must be drawn accurately and neatly.
6. ALL the questions must be answered on the QUESTION PAPER, as instructed.
7. ALL the pages, irrespective of whether the question was attempted or not, must be re-stapled in numerical sequence in the TOP LEFT-HAND CORNER ONLY.
8. Time management is essential in order to complete all the questions.
9. Print your examination number in the block provided on every page.
10. Any details or dimensions not given must be assumed in good proportion.

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|-----------------------|----------------|---|---|---------------|------|-----------|---|---|---------------|------|------------|
| QUESTION              | MARKS OBTAINED |   |   | $\frac{1}{2}$ | SIGN | MODERATED |   |   | $\frac{1}{2}$ | SIGN | RE-MARKING |
| 1                     |                |   |   |               |      |           |   |   |               |      |            |
| 2                     |                |   |   |               |      |           |   |   |               |      |            |
| 3                     |                |   |   |               |      |           |   |   |               |      |            |
| 4                     |                |   |   |               |      |           |   |   |               |      |            |
| TOTAL                 |                |   |   |               |      |           |   |   |               |      |            |
|                       | 2              | 0 | 0 |               |      | 2         | 0 | 0 |               |      | 2 0 0      |

FINAL CONVERTED MARK

100

CHECKED BY

### COMPLETE THE FOLLOWING:

CENTRE NUMBER

CENTRE NUMBER

EXAMINATION NUMBER

EXAMINATION NUMBER

**WELDING SYMBOL**

M 25 - 75 TIG

| PARTS LIST |                    |          |            |
|------------|--------------------|----------|------------|
| NO.        | PARTS              | QUANTITY | MATERIAL   |
| 1          | BASE PLATE         | 1        | CAST IRON  |
| 2          | CRADLE             | 1        | CAST IRON  |
| 3          | SWIVEL JAW         | 1        | CAST IRON  |
| 4          | SLIDING JAW        | 1        | CAST IRON  |
| 5          | JAW PLATES         | 2        | MILD STEEL |
| 6          | M10 BOLT           | 2        | CAST IRON  |
| 7          | M12 NUT            | 2        | TOOL STEEL |
| 8          | ELLIPTICAL BRACKET | 1        | CAST IRON  |

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HEAT TREATMENT:  
TEMPERING AT 860°  
WITH INDUCTION  
HEATING

UNSPECIFIED RADII ARE R2.

**SWIVEL VICE**

FILE NAME: KRJM314

DRAWING No. CLMP02

TOLERANCE: +0,23  
-0,1

DRAWING PROGRAM: AUTOCAD 2024

SCALE 1 : 3

DRAWN BY: ALISTER

CHECKED BY: ERIC

APPROVED BY: SAMANTHA

DATE: 13/07/2024

DATE: 14/08/2024

DATE: 05/09/2024

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UNSPECIFIED RADII ARE R2.

**SWIVEL VICE**

HEAT TREATMENT:  
TEMPERING AT 860°  
WITH INDUCTION  
HEATING

UNSPECIFIED RADII ARE R2.

**SWIVEL VICE**

**QUESTION 1: ANALYTICAL (MECHANICAL)**

**Given:**  
Two views of a swivel vice, a welding symbol, a parts list, a title block and a table of questions. The drawing is not presented to the indicated scale.

**Instructions:**  
Complete the table below by neatly answering the questions, which refer to the accompanying drawing, title block and mechanical content.

| QUESTIONS |                                                                                                                                                                                        | ANSWERS              |        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------|
| 1         | Who approved the drawing?                                                                                                                                                              | 1                    |        |
| 2         | What is the name of the company?                                                                                                                                                       | 1                    |        |
| 3         | Which drawing program was used to prepare the drawing?                                                                                                                                 | 1                    |        |
| 4         | What projection system is indicated by the symbol in the title panel?                                                                                                                  | 1                    |        |
| 5         | From what material are the jaw plates made?                                                                                                                                            | 1                    |        |
| 6         | How many M10 bolts are there in a single swivel vice assembly?                                                                                                                         | 1                    |        |
| 7         | At what temperature must the tempering be done?                                                                                                                                        | 1                    |        |
| 8         | If VIEW 1 is the front view, what would VIEW 2 be called?                                                                                                                              | 1                    |        |
| 9         | What is the correct label for VIEW 3?                                                                                                                                                  | 1                    |        |
| 10        | Determine the complete dimensions at: A: B: C: D: E:                                                                                                                                   | 5                    |        |
| 11        | Name the feature at F.                                                                                                                                                                 | 1                    |        |
| 12        | Name the feature at G.                                                                                                                                                                 | 1                    |        |
| 13        | If a scale of 1 : 1 was used, how would the dimension at H read?                                                                                                                       | 1                    |        |
| 14        | What type of section is shown at J?                                                                                                                                                    | 1                    |        |
| 15        | What type of section is shown at K?                                                                                                                                                    | 1                    |        |
| 16        | With reference to the welding symbol, what does M indicate?                                                                                                                            | 1                    |        |
| 17        | With reference to the welding symbol, what does '25 - 75' indicate?                                                                                                                    | 25:<br>75:           | 1<br>1 |
| 18        | With reference to the tolerance, calculate the <b>maximum</b> and the <b>minimum</b> size of the dimension at S.                                                                       | MAXIMUM:<br>MINIMUM: | 1<br>1 |
| 19        | In the space below (ANSWER 19), complete, in neat freehand, the sectional view of the given mechanical part according to cutting plane A-A.                                            |                      | 3      |
| 20        | With reference to the machining symbol in the space below (ANSWER 20), ONLY add the letters A, B and C to the symbol, in the correct position, that describe the given specifications. |                      | 3      |
| TOTAL     |                                                                                                                                                                                        | 30                   |        |

ANSWER 19

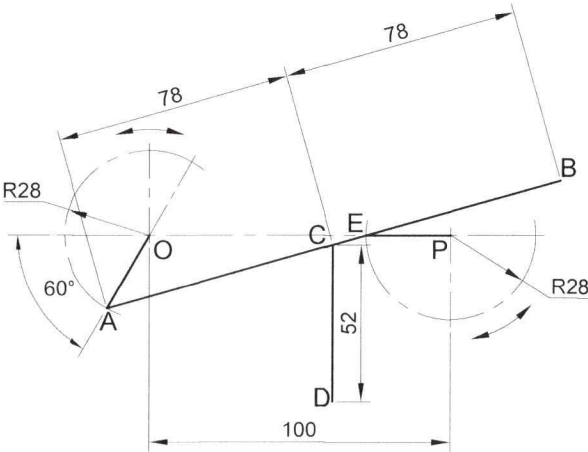
ANSWER 20

A = DIRECTION OF LAY  
B = ROUGHNESS VALUE  
C = METHOD

EXAMINATION NUMBER

EXAMINATION NUMBER

STAPLE



QUESTION 2: LOCI

NOTE: Answer QUESTIONS 2.1 and 2.2.

2.1 MECHANISM

- Given:**
- A schematic drawing of a mechanism consisting of crank OA, slotted link AB, rod CD and crank PE
  - The position of centre point O on the drawing sheet
- Specifications:**
- Crank OA is pin-jointed to slotted link AB at A.
  - End E of crank PE slides along the slot of link AB.
  - Rod CD is pin-jointed to slotted link AB at C.

**Motion:**  
As crank OA starts its oscillation in a clockwise direction through 180° around centre point O, crank PE starts its oscillation at the same speed through 180° in an anti-clockwise direction around centre point P. During the movement, end E of crank PE reciprocates in the slot of link AB. Rod CD remains vertical throughout the movement of the two cranks.

- Instructions:**
- Using centre point O, draw, to scale 1 : 1, the given schematic drawing, including the labels.
  - Trace the loci generated by point B and point D for ONE oscillation.
  - Show ALL construction. [20 1/2]

| ASSESSMENT CRITERIA 2.1 |              |        |  |  |
|-------------------------|--------------|--------|--|--|
| 1                       | GIVEN + CL   | 7 1/2  |  |  |
| 2                       | CONSTRUCTION | 4      |  |  |
| 3                       | LOCUS OF B   | 4 1/2  |  |  |
| 4                       | LOCUS OF D   | 4 1/2  |  |  |
| PENALTIES (-)           |              |        |  |  |
| SUBTOTAL                |              | 20 1/2 |  |  |



ANSWER  
MOTION FROM 135° TO 225°: \_\_\_\_\_

2.2: CAM

- Given:**
- The detail of a camshaft and a roller-follower at the minimum distance from the camshaft centre
  - The displacement graph, correctly aligned for the given follower and camshaft
- Specifications:**
- The roller-follower reciprocates along the 30° centre line that passes through the centre of the camshaft.
  - Roller = Ø12
  - Rotation = clockwise

- Instructions:**
- Project and draw the cam profile from the given displacement graph.
  - Indicate the direction of rotation on the cam profile with an arrow.
  - Show ALL necessary construction and projection.
  - In the space provided, name the type of motion from 135° to 225°. [19 1/2]

| ASSESSMENT CRITERIA 2.2 |                |        |  |  |
|-------------------------|----------------|--------|--|--|
| 1                       | CONSTRUCTION   | 5      |  |  |
| 2                       | CAM PROFILE    | 13 1/2 |  |  |
| 3                       | TYPE OF MOTION | 1      |  |  |
| PENALTIES (-)           |                |        |  |  |
| SUBTOTAL 2.2            |                | 19 1/2 |  |  |
| SUBTOTAL 2.1            |                | 20 1/2 |  |  |
| TOTAL                   |                | 40     |  |  |

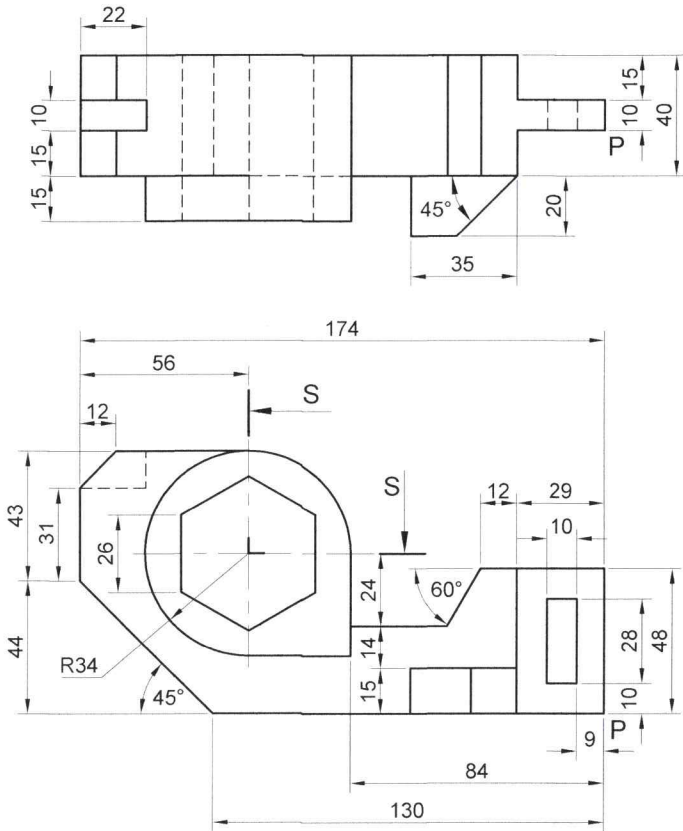
|                    |   |
|--------------------|---|
| EXAMINATION NUMBER |   |
| EXAMINATION NUMBER | 3 |



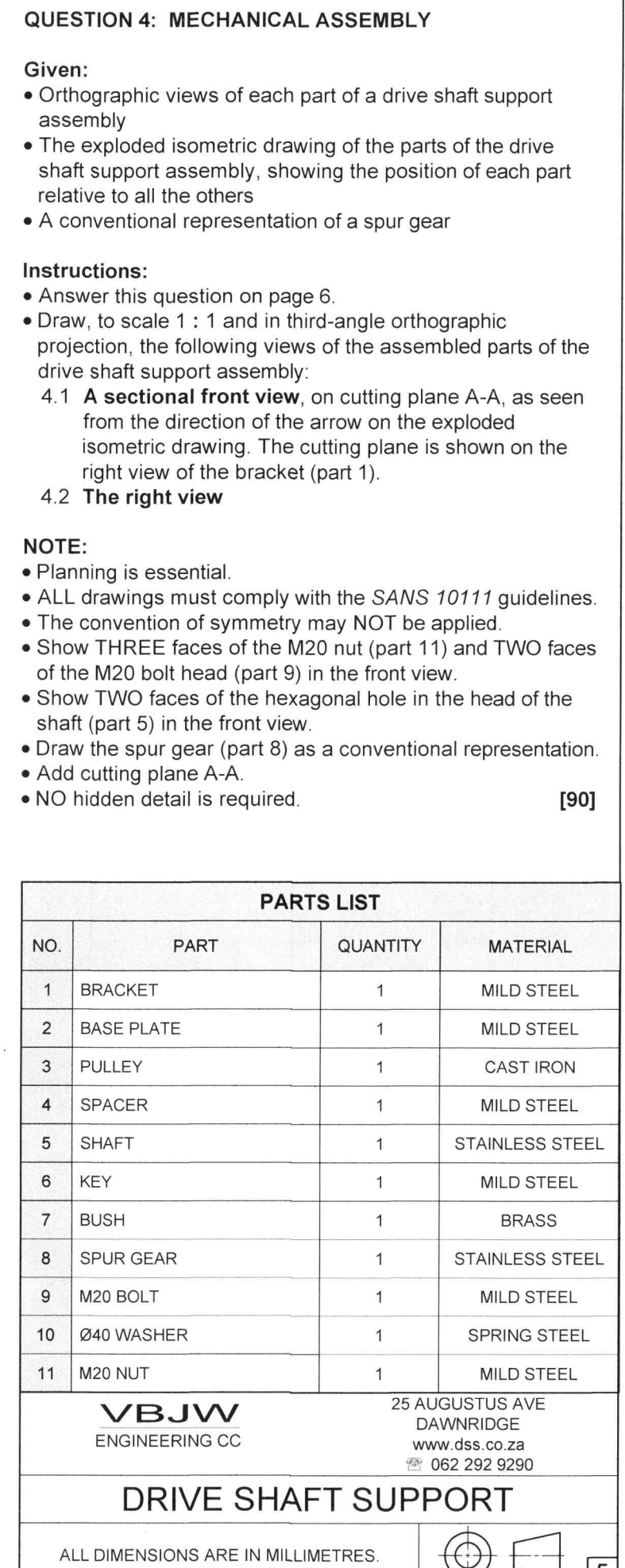
QUESTION 3: ISOMETRIC DRAWING

- Given:**
- The front view and top view of a gauge block
  - The position of point P on the drawing sheet

- Instructions:**
- Using scale 1 : 1, convert the orthographic views of the gauge block into a sectional isometric drawing on cutting plane S-S.
- Use P as the starting point and lowest point of the drawing.
  - Show ALL construction.
  - NO hidden detail is required.
- [40]



| ASSESSMENT CRITERIA |                                  |        |  |   |
|---------------------|----------------------------------|--------|--|---|
| 1                   | PLACING/ORIENTATION + AUX. VIEWS | 2      |  |   |
| 2                   | FRONT PORTION                    | 16 1/2 |  |   |
| 3                   | MIDDLE + REAR PORTION            | 7      |  |   |
| 4                   | HEXAGON                          | 4 1/2  |  |   |
| 5                   | CIRCLE + CL                      | 5      |  |   |
| 6                   | SECTIONED SURFACE + HATCHING     | 5      |  |   |
| PENALTIES (-)       |                                  |        |  |   |
| TOTAL               |                                  | 40     |  |   |
| EXAMINATION NUMBER  |                                  |        |  |   |
| EXAMINATION NUMBER  |                                  |        |  | 4 |



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| INCORRECT ORTHOGRAPHIC PROJECTION |  |
| INCORRECT OVERALL SCALE           |  |
| INCORRECT HATCHING                |  |
| PARTS NOT ASSEMBLED               |  |
|                                   |  |
| TOTAL PENALTIES (-)               |  |

| ASSESSMENT CRITERIA  |                         |                                |          |      |           |
|----------------------|-------------------------|--------------------------------|----------|------|-----------|
| RIGHT VIEW           |                         |                                |          |      |           |
|                      |                         | POSSIBLE                       | OBTAINED | SIGN | MODERATED |
| 1                    | BRACKET +<br>BASE PLATE | 5 <sup>1</sup> / <sub>2</sub>  |          |      |           |
| 2                    | GEAR + CL               | 2                              |          |      |           |
| 3                    | PULLEY +<br>SHAFT       | 2                              |          |      |           |
| SUBTOTAL             |                         | 9 <sup>1</sup> / <sub>2</sub>  |          |      |           |
| SECTIONAL FRONT VIEW |                         |                                |          |      |           |
| 1                    | BASE                    | 4                              |          |      |           |
| 2                    | BRACKET + RIB           | 10 <sup>1</sup> / <sub>2</sub> |          |      |           |
| 3                    | PULLEY                  | 10                             |          |      |           |
| 4                    | SHAFT                   | 11                             |          |      |           |
| 5                    | BUSH + KEY +<br>SPACER  | 6 <sup>1</sup> / <sub>2</sub>  |          |      |           |
| 6                    | SPUR GEAR               | 11                             |          |      |           |
| 7                    | M20 NUT +<br>WASHER     | 5                              |          |      |           |
| 8                    | M20 BOLT                | 7 <sup>1</sup> / <sub>2</sub>  |          |      |           |
| SUBTOTAL             |                         | 65 <sup>1</sup> / <sub>2</sub> |          |      |           |
| GENERAL              |                         |                                |          |      |           |
| 1                    | CENTRE LINES            | 2                              |          |      |           |
| 2                    | CUTTING<br>PLANE        | 3                              |          |      |           |
| 3                    | ASSEMBLY                | 10                             |          |      |           |
| SUBTOTAL             |                         | 15                             |          |      |           |
| TOTAL                |                         | 90                             |          |      |           |
| PENALTIES (-)        |                         |                                |          |      |           |
| GRAND TOTAL          |                         |                                |          |      |           |
| EXAMINATION NUMBER   |                         |                                |          |      |           |
|                      |                         |                                |          |      |           |
| EXAMINATION NUMBER   |                         |                                |          |      | 6         |